



Community and International Nutrition

Preconception Micronutrient Supplementation Affects Maternal BMI and Body Composition Postpartum: A Randomized Controlled Trial in Vietnam^{☆,☆☆}

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ABSTRACT

Background: Although there is growing evidence on the role of preconception nutrition for birth outcomes, limited evidence exists for its effects on maternal health.

Objectives: This study evaluates the impact of preconception micronutrient supplementation on maternal BMI (kg/m²) and body composition at 6 to 7 y postpartum (PP).

Methods: We followed females who participated in a randomized controlled trial of preconception supplementation in Vietnam and delivered live offspring ($n = 1599$). Females received weekly supplements containing either 2800 µg folic acid (FA) only, 60 mg iron and 2800 µg FA (IFA), or multiple micronutrients (MMs) (15 micronutrients including IFA) from baseline until conception followed by daily prenatal IFA supplements until delivery. Height, weight, mid-upper arm circumference, triceps skinfold, and waist-hip circumference were measured at recruitment and at 1, 2, and 6 to 7 y PP. Body fat was assessed using bioelectric impedance at 6 to 7 y PP ($n = 867$). Group comparisons were made using analysis of variance or chi-square tests and general linear models for adjusted models.

Results: At 6 to 7 y PP, we found significant differences ($P < 0.05$) by treatment group for mean percent fat (MM: 29.2%; IFA: 27.6%; FA: 27.8%), absolute fat mass (MM: 15.1 kg; IFA: 14.0 kg; FA: 14.3 kg), and prevalence of underweight based on BMI < 18.5 (MM: 5.8%; IFA: 10.3%; FA: 14.3%). Mean BMI and triceps skinfold thickness were higher in the MM group, but these differences were not statistically significant; the differences in absolute fat mass were also attenuated after controlling for body weight. No differences were observed for fat-free mass, prevalence of overweight (BMI > 23), or other anthropometric measurements.

Conclusions: Preconception MM supplementation was associated with lower prevalence of underweight and higher percent fat when compared with IFA and/or FA only. Preconception micronutrient interventions may have long-term effects on maternal health and merit further examination.

This trial was registered at clinicaltrials.gov as [NCT01665378](https://clinicaltrials.gov/ct2/show/study?term=NCT01665378).

Keywords: preconception, multiple micronutrient, maternal body composition, randomized controlled trial, Vietnam

Abbreviations: ANOVA, analysis of variance; BIA, bioelectric impedance analysis; FA, folic acid; IFA, iron and folic acid; MM, multiple micronutrient; MUAC, mid-upper arm circumference; NCD, noncommunicable disease; PP, postpartum; RCT, randomized controlled trial; WRA, women of reproductive age.

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Introduction

Pregnancy is a period of rapid physical and physiological change during which nutrient requirements increase to meet metabolic demands, promote adequate weight gain, and support fetal growth and development [1]. Many women of reproductive age (WRA) in developing countries, however, often have micronutrient-poor diets and enter pregnancy malnourished [2]. The heightened nutritional demands during pregnancy further exacerbate existing micronutrient deficiencies and contribute to poor health outcomes for both mother and child [3]. Evidence from several randomized controlled trials (RCTs) supports the benefits of multiple micronutrient (MM) supplementation during pregnancy for offspring health, including reductions in low birth weight and small-for-gestational age [4,5]. There is also increased interest in optimizing nutritional status among WRA prior to conception, especially because many females in resource-poor settings often do not receive prenatal care until midpregnancy [6].

Maternal preconception nutritional status may have lasting impacts for both the mother and her offspring [7]. Maternal undernutrition before pregnancy (being underweight or having micronutrient deficiencies) is associated with several adverse birth outcomes including low birth weight, neural tube defects, and preterm delivery [8,9]. In parallel, maternal overnutrition is associated with adverse birth outcomes, especially preterm birth, increased risk of cardiometabolic disorders both during pregnancy (e.g., preeclampsia, gestational diabetes) and after pregnancy (e.g., type 2 diabetes, cardiovascular disease), and an increased risk of childhood obesity [10,11]. Furthermore, maternal preconception BMI has been positively associated with gestational weight gain and maternal and child percent body fat at 6 to 7 y postpartum (PP) [12]. Thus, the preconception period offers a valuable opportunity to prevent intergenerational increases in noncommunicable diseases (NCDs), especially within a changing nutrition environment.

Although inadequate food intakes and food insecurity continue to affect many communities especially in low- and middle-income countries, global urbanization, changes in the food supply and diet patterns, and reductions in physical activity have contributed to increases in the prevalence of overweight and obesity among WRA worldwide [13]. Recent estimates suggest that nearly 21% of the world's females will be obese by 2025 [14]. Furthermore, the rate at which obesity is increasing in low- and middle-income countries is faster than that observed in higher-income countries; subsequently, these populations may be at higher risk of NCDs including type 2 diabetes and cardiovascular disease [15]. In Vietnam, despite significant declines in the past 2 decades, nearly 1 in 5 WRA are underweight (BMI < 18.5 kg/m²) [16] while the prevalence of obesity and several NCDs is increasing [17–19].

To date, most preconception and prenatal nutrition trials have assessed offspring birth outcomes, yet very little is known about the impact of nutrition interventions prior to conception on long-term maternal health and well-being [4,20,21]. The few maternal outcomes that have been assessed are limited to complications during pregnancy, such as anemia, preeclampsia, and c-sections, along with maternal mortality [4]. Among WRA, excess iron stores may be associated with higher risk of cardiovascular disease and type 2 diabetes [22,23]. Some evidence

suggests that micronutrients like iron and zinc may increase appetite and thereby food intakes [24]. Providing MMs before and during pregnancy may therefore contribute to increased energy intake during pregnancy, increased weight gain during pregnancy, and possibly increases in PP weight retention and changes in body composition. However, few studies have followed up females for later outcomes. Findings from 1 RCT in Mexico suggest that prenatal MM supplements may contribute to increased weight retention at 1 mo PP among overweight females [25]. To our knowledge, no intervention studies have assessed the long-term effects on maternal nutritional status using anthropometric indicators such as BMI or maternal body composition.

We conducted a randomized controlled trial (PRECONCEPT) in rural Vietnam where young WRA were randomized to receive weekly supplements containing only folic acid (FA), iron and folic acid (IFA), and MMs [26]. Nearly a third of the females were underweight at baseline and had suboptimal intakes of several micronutrients [27,28]. All females who conceived received daily IFA through delivery and were followed up for birth outcomes and through 6 to 7 y PP for offspring growth and development [29]. The objective of the current study is to fill the research gap on maternal health outcomes by evaluating the impact of preconception weekly MMs on maternal body composition and nutritional status, including BMI, mid-upper arm circumference (MUAC), skinfold thickness, and waist-to-hip ratio at 6 to 7 y PP in Vietnam.

Methods

Study design, participants, and setting

This study used data from an RCT (PRECONCEPT study) conducted in 20 communes of Vietnam from 2011–2016 that evaluated the effects of preconception micronutrient supplementation on maternal and child health outcomes [26]. Details on the trial design have been published previously [26]. In brief, a total of 5011 WRA were recruited and were randomly assigned to receive weekly supplements containing either 1) 2800 µg FA only; 2) 60 mg iron and 2800 µg FA (IFA), or 3) MMs including 60 mg iron, 2800 µg FA, 600 IU of vitamin D, and 1 to 2 recommended dietary allowances of 12 other vitamins (A, B₁, B₂, B₃, B₆, B₁₂, C, E) and minerals (Zn, Cu, Se, I₂) from baseline until conception. The micronutrient composition of the supplements is shown in Supplemental Table 1. During pregnancy, all females received daily IFA (60 mg iron and 400 µg FA) as standard of care through delivery. A total of 1813 females became pregnant, contributing to 1639 live births, of which 1599 had known birth outcomes and were followed up through age 2 y. The next follow-up was conducted in 2019–2020 when the offspring were 6 to 7 y of age [28]. We updated contact information and invited all study participants who lived in the study area to come for in-person visits to the Community Health Center for all follow-up maternal and child assessments. Due to budgetary limitations, we were unable to follow-up households who had migrated or left the study area but had planned to obtain data on ≥90% of the original mother–child pairs ($n = 1440$), which would have ≥90% power to detect effect sizes of ≥0.3 SD for key outcome measures [30]. The subsample of females with anthropometric measurements at baseline and 6 to 7 y PP were eligible for this analysis.

Outcome measures

Maternal anthropometric measurements were obtained at recruitment and at 1 y, 2 y, and 6 to 7 y PP by trained staff using standard procedures [31,32]. All measurements were taken twice with strict adherence to the measuring techniques and recording procedures. Weight was measured using calibrated electronic Seca scales to the nearest 0.10 kg. Standing height was measured using a portable stadiometer, measured to the nearest centimeter. BMI was calculated, and females were categorized as underweight (<18.5), normal ($18.5 < \text{BMI} < 23$), or overweight (≥ 23) using cutoff values recommended to identify under- and overnutrition in Asians [33]. Other anthropometric measures included mid-upper arm, waist and hip circumferences, and triceps skinfolds as proxy measures of body fat (total and distribution). All anthropometrists were blinded to the original intervention assignments.

Maternal body composition was measured at 6 to 7 y PP using a Seca mBCA 525 multifrequency bioelectric impedance analysis (BIA, Seca Corporation). All subjects had fasted for ≥ 4 h with no vigorous exercise on the day before the measurement. On the test day, females lay supine on a table made of nonconductive materials, and the measurements were performed using a standardized procedure [34]. Briefly, the females wore minimal clothing and rested in the prone position for 5 min prior to the measurement. Each female's arms and legs were positioned according to the manufacturer's protocol. Using an 8-polar tactile-electrode impedance meter, 4 electrodes were placed on the palm and thumb of both hands, and 4 electrodes were placed on the anterior and posterior aspects of the soles of both feet. Bioelectrical impedance was measured at multiple frequencies (1, 2, 5, 10, 20, 50, 100, 200, 500 kHz), and body composition was determined using the manufacturer's algorithm. The main outcomes used for the analyses included absolute fat mass (kilograms), fat-free mass (kilograms), and percentage total body fat.

Other covariates

At the time of enrollment (preconception), we obtained data on demographic characteristics (age, ethnicity, education, and occupation), household socioeconomic status, and food security. Using multiple variables including household ownership of different assets, livestock, house and land, and key housing characteristics, we constructed a household socioeconomic status index by performing principal component analysis and dividing the first component into tertiles [35,36]. We also constructed a household food security score using responses to 9 items related to the household's experience of food insecurity in the past 30 d [37] and classified individuals as food secure or insecure.

The duration of exposure was calculated as the total number of weeks that the females received the supplements from preconception to the confirmation of pregnancy when they switched to daily IFA. All supplements were distributed biweekly by trained village health workers who monitored the consumption by direct observation during their visit and weekly recall by telephone. We also measured compliance as a proportion of the total number of weeks that they should have received the intervention [38]. Data on child feeding practices were obtained from the mothers during the 1 and 2 y PP follow-up visits as well as on additional births, dietary intakes, and physical activity at 6

to 7 y PP. Dietary diversity was assessed by asking all foods and beverages consumed in the past 24 h. All foods reported were then grouped into 10 categories based on the Minimum Dietary Diversity Guidelines for Women [39]. Physical activity patterns were measured using the Global Physical Activity Questionnaire [40], which is validated in several countries including Vietnam [41–43]. Physical activity of all females was determined by estimating the time spent on various work, transport, and leisure time activities and multiplying by the appropriate metabolic equivalents [34].

Statistical analysis

Proportions, means, and SDs are presented for selected maternal characteristics at enrollment and follow-up by treatment group for the final analytic sample. We also compared baseline characteristics of study participants in the final analytic sample and those lost to follow-up and used analysis of variance (ANOVA) and chi-square tests to test group differences in means and proportions, respectively.

We first followed the intention-to-treat approach to assess the effect of treatment on maternal anthropometric indicators at follow-up and used ANOVA and chi-square tests to test unadjusted differences in means and proportions, respectively, between the 3 treatment groups. We also examined differences in percentage body fat, absolute body fat mass, and fat-free mass by treatment group using ANOVA and evaluated effect modification by testing interactions between treatment group and baseline BMI that were specified a priori. We used $P < 0.05$ for all tests of main effects and applied Bonferroni corrections for multiple comparisons testing and accounting for the fact that the outcome measures were related to each other. In addition to intent-to-treat analysis, we also examined models that adjusted for age at follow-up, duration of intervention, and total body weight (only for absolute fat and fat-free mass) as well as other covariates (having another child during follow-up, gestational weight gain, and exclusive breastfeeding) using general linear models.

All reported P values were from 2-tailed tests. Data were analyzed with STATA software version 16.0 (Stata Corporation).

Ethical approval

The study was approved by the Ethical Committee of Thai Nguyen National Hospital in Vietnam and Emory University's Institutional Review Board, Atlanta, GA, United States. The trial is registered in the United States Clinical Trials registry (NCT01665378). Written informed consent was obtained from all study participants.

Results

Study sample

The final analytical sample included all females who were randomized to treatment, had live births, were not pregnant at the time of follow-up, and had weight measurements available at 6 to 7 y PP ($n = 867$), as shown in Figure 1. Although we had contacted over 90% of the original households for this follow-up ($n = 1440$), several mothers did not attend the study visit primarily due to: 1) lack of time (466 mothers sought work in large towns or cities and had left their children in the village with

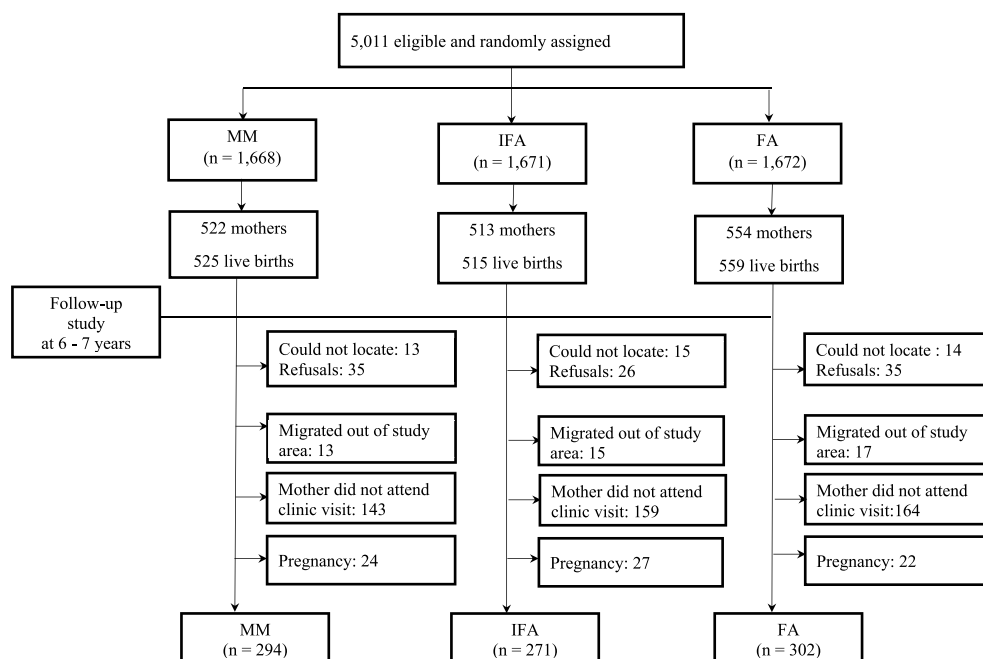


FIGURE 1. Flow of sample selection for the PRECONCEPT trial. FA, folic acid; IFA, iron and folic acid; MM, multiple micronutrient.

grandparents, relatives, etc.); 2) 45 had migrated from the study area; and 3) 96 refused to participate. Although the analytic sample was similar on most baseline characteristics to those missing data and did not differ by treatment group, mothers included in the analytic sample were older (26.4 y compared with 25.3 y, $P < 0.001$) and more likely to have had another child during the follow-up period (18.3% compared with 5.2%, $P < 0.001$) (Supplemental Table 2).

Maternal and household characteristics by treatment group

Table 1 shows selected maternal and household characteristics measured at the time of study enrollment (preconception) and follow-up by treatment group in the analytic sample. At enrollment, the mean age of females was ~26 y, and >90% had ≥ 1 child. Nearly one-third of the study population was underweight at preconception, and mean gestational weight gain was 10.2 kg. As previously reported, compliance was >90% for both preconception and prenatal supplements and did not differ by treatment group [44]. Overall, ~60% of females were exclusively breastfeeding their children at 6 mo PP, and nearly 96% of females continued to breastfeed at 1 y PP. There were no differences by treatment group for key maternal characteristics at enrollment or during the follow-up period, including dietary diversity and physical activity at 6 to 7 y PP.

Maternal anthropometry and body composition at 6–7 y PP

Descriptive statistics for maternal anthropometry and body composition measured at 6 to 7 y PP by treatment group are presented in Table 2. Maternal BMI and percent body fat differed by treatment group, but after adjusting for multiple comparisons only the differences in mean (SD) percent body fat remained statistically significant [MM: 29.2 (5.7); IFA: 27.6 (6.1); FA: 27.8 (6.7)]. The differences in mean percent body fat remained

statistically significant after adjusting for age and duration of the intervention as well as weight gain during pregnancy, exclusive breastfeeding, and having another child during the follow-up period. Although absolute fat mass was higher in the MM group compared to the IFA and FA groups and these differences remained statistically significant after adjusting for age, duration of intervention, and having another child during the follow-up period, the results were attenuated when controlling for total body weight. There were no differences by treatment group in absolute fat-free mass, MUAC, and waist-to-hip ratio. Although mean triceps skinfold thickness was higher in the MM group, the differences were not statistically significant.

The overall mean (SD) BMI increased from 19.7 (2.1) at baseline to 21.6 (2.7) at 6 to 7 y PP. The prevalence of overweight also increased from 7.2% to 24.9% from baseline to 6 to 7 y PP, whereas underweight declined from 31.1% to 10.2%, respectively. There were significant differences in the prevalence of maternal underweight by treatment group, with no differences for overweight at 6 to 7 y PP (Figure 2). Finally, we also found that females who were underweight at baseline and received MMs had higher body fat percentage compared with IFA and FA only ($P < 0.05$), with no significant differences by treatment group for the females who were normal or overweight at baseline (Figure 3). These differences also remained after adjusting for age, duration of intervention, and other covariates including gestational weight gain, exclusive breastfeeding, and having another child during the follow-up period.

Discussion

The prevalence of micronutrient-poor diets among WRA continues to remain high across the globe [2]. Moreover, the increased need for nutrients during pregnancy can exacerbate existing micronutrient deficiencies and contribute to poor fetal development and poor PP maternal health and nutritional status

TABLE 1Comparison of maternal and household characteristics at study enrollment (preconception) and follow-up by treatment group^{1,2}

	MM (n = 294)	IFA (n = 271)	FA (n = 302)
Maternal characteristics at enrollment			
Age, y	26.6 ± 4.7	26.7 ± 4.3	26.1 ± 4.2
Ethnic minority, %	51.7	47.2	51.0
Education level, %			
Primary school	8.8	7.7	5.6
Secondary school	52.4	50.9	57.3
High school	27.2	25.5	22.8
College or higher	11.6	15.9	14.2
Work as farmers, %	80.3	77.1	77.5
Having ≥1 child, %	94.0	94.9	95.1
Number of children, n	2.0 ± 0.4	2.0 ± 0.3	2.0 ± 0.4
Anthropometry at enrollment			
Weight, kg	46.1 ± 5.9	45.7 ± 5.6	45.9 ± 5.8
Height, cm	152.7 ± 5.2	152.6 ± 5.3	153.0 ± 5.1
MUAC, cm	24.8 ± 2.4	24.5 ± 2.2	24.6 ± 2.4
BMI, kg/m ²	19.8 ± 2.1	19.6 ± 2.0	19.6 ± 2.3
Underweight (BMI <18.5 kg/m ²), %	30.0	29.9	33.3
Normal weight (18.5 ≤ BMI <23 kg/m ²), %	63.5	63.5	58.3
Overweight (BMI ≥23 kg/m ²), %	6.5	6.6	8.3
Household characteristics at enrollment			
Socio-economic status ³ , %			
Low	35.4	34.3	31.6
Average	32.3	32.1	35.2
High	32.3	33.6	33.2
Food security ⁴ , %	72.1	75.3	70.7
Follow-up			
Duration of intervention, wk	33.5 ± 25.1	33.2 ± 24.4	33.5 ± 24.0
Received supplements ≥26 wk, %	50.7	50.6	52.5
Gestational weight gain, kg	10.0 ± 3.9	10.5 ± 4.1	10.2 ± 3.9
Having another child at follow-up, %	18.4	18.8	17.9
Exclusive breastfeeding at 6 mo, %	59.7	55.5	64.8
Breastfeeding at 12 mo, %	94.5	96.8	96.5
Breastfeeding at 24 mo, %	3.8	4.7	3.8
Number of food groups consumed at 6–7 y, n	5.1 ± 1.2	5.1 ± 1.1	5.1 ± 1.3
Physical activity at 6–7 y, MET, median (IQR)	8657 (2226–14,202)	8648 (2687–14,880)	8957 (2413–14,826)

Abbreviations: ANOVA, analysis of variance; FA, folic acid; IFA, iron and folic acid; MET, metabolic equivalents; MM, multiple micronutrient; MUAC, mid-upper arm circumference.

¹ ANOVA and chi-square tests were used to test differences in means and proportions, respectively, between groups. There were no significant differences by treatment group for any of the characteristics included in the table ($P > 0.05$).

² Values are mean ± SD or percentage.

³ Household socioeconomic status index was constructed via principal component analysis using multiple variables including household ownership of different assets, livestock, house and land, and key housing characteristics and was subsequently divided into tertiles.

⁴ Household food security index was constructed using 9 items related to household's access to food in past 30 d and categorized as "food secure" or "food insecure".

[3]. Although there have been a few MM trials prior to conception [21,45–47], data are limited on the long-term impact on the health and nutritional status of the mother. Briefly, in this large RCT of weekly MM supplementation prior to conception, compared with IFA and FA only, there was a lower prevalence of underweight and higher mean percentage body fat in the MM group at 6 to 7 y PP. Mean BMI and absolute fat mass were also higher in the MM group compared with IFA and FA, with no differences in triceps skinfold thickness, fat-free mass, MUAC, or waist-to-hip ratio. Finally, females in the MM group who were underweight at baseline had higher percentage body fat when compared with the other groups, with no differences among those who were normal or overweight at baseline.

Several plausible factors may explain these findings. First, improving maternal nutritional status prior to conception may affect appetite during pregnancy. Micronutrients like iron and zinc may improve appetite [24,48], and possibly, females who

received MMs prior to conception may have entered pregnancy with an increased appetite, facilitating appropriate gestational weight gain and a healthy PP weight. Although we did not observe differences in gestational weight gain by treatment, we did not measure nutrient intakes or appetite during pregnancy, which could have been affected during gestation and persisted PP. Some evidence also suggests that preconception MM supplementation may influence energy metabolism and inflammation during pregnancy. For example, 1 study showed that MM supplementation during pregnancy was associated with post-supplementation reductions in C-reactive protein and serum leptin [49]. Although we do not know if these changes remain over time, reductions in key hormones such as serum leptin may be sustained and could have long-term effects on energy balance through its effects on appetite and decreasing energy expenditure. In contrast, findings from a trial in India showed that provision of a micronutrient-rich snack prior to and during

TABLE 2
Comparison of maternal BMI, body composition, and anthropometry at 6 to 7 y PP by treatment group¹

	MM (n = 294)	IFA (n = 271)	FA (n = 302)	P value ²
BMI, kg/m ²	21.9 ± 2.7	21.4 ± 2.6	21.4 ± 2.9	0.035
Percent fat, %	29.2 ± 5.7	27.6 ± 6.1	27.8 ± 6.7	0.004 ^{3,4}
Absolute fat mass, kg	15.1 ± 4.7	14.0 ± 4.8	14.3 ± 5.0	0.016 ^{3,5}
Fat-free mass, kg	35.8 ± 3.8	35.7 ± 3.5	35.8 ± 3.5	0.84 ⁵
MUAC, cm	26.8 ± 2.7	26.4 ± 2.7	26.5 ± 2.9	0.19
Triceps skinfold, mm	16.4 ± 5.9	15.6 ± 5.5	15.8 ± 5.7	0.18
Waist-to-hip ratio, <i>n</i>	0.9 ± 0.1	0.9 ± 0.1	0.9 ± 0.1	0.77
Weight, kg	50.9 ± 7.2	49.7 ± 7.0	50.0 ± 7.1	0.13
Height, cm	152.4 ± 5.6	152.5 ± 5.2	152.9 ± 5.1	0.50

Abbreviations: ANOVA, analysis of variance; FA, folic acid; IFA, iron and folic acid; MM, multiple micronutrient; MUAC, mid-upper arm circumference; PP, postpartum; SD, standard deviation.

¹ Values are mean ± SD.

² ANOVA was used to assess differences in anthropometric indicators and measures of body composition between treatment groups for intention-to-treat analysis.

³ $P < 0.05$ using general models that adjusted for age, duration of intervention, and having another child during the follow-up period.

⁴ $P < 0.05$ using general linear models that adjusted for age, duration of intervention, having another child, gestational weight gain, and exclusive breastfeeding.

⁵ $0.05 < P \leq 0.1$ using general linear models that adjusted for weight with or without additional covariates.

pregnancy prevented gestational diabetes mellitus, adjusting for prepregnancy adiposity and gestational weight gain [50].

To fully appreciate the results presented, it is important to consider the impact of several important covariates that may be associated with maternal BMI and body composition at 6 to 7 y PP, including body size, gestational weight gain, breastfeeding status and duration, physical activity (quantified by median metabolic equivalents), dietary diversity (number of food groups consumed), and percent of females who had another child at follow-up. Some evidence suggests that breastfeeding practices may influence PP weight change [51]. However, there were no differences in breastfeeding practices including exclusive breastfeeding by treatment groups. Similarly, there were also no differences by treatment group for measures of physical activity, dietary diversity, and parity at follow-up. Notably, over 90% of females already had 1 child upon enrollment in the study, and fewer than 20% had subsequent pregnancies during the

follow-up period with no differences by treatment group. These numbers are in accordance with the Vietnamese family planning efforts that discourage families from having >2 children. Age may also contribute to differences in maternal PP body composition; however, there was no major variation in the age of mothers across treatment groups in our trial. Furthermore, although there were significant differences between the groups for mean percent body fat, there were no differences when absolute fat mass was adjusted for total body weight. This is not an unexpected outcome because percent body fat is a simple, but not completely appropriate means to compare differences in body fat mass. Given that the ratio of body fat mass to total body weight is neither linear nor crosses a 0-intercept, it is necessary to adjust fat mass for total body weight using multiple linear regression analysis to normalize fat mass for body size [52]. Regardless, the higher fat mass in the MM group remains a concern and emphasizes the need to have better methods to assess body composition beyond anthropometric measurements among WRA. Further, our findings remained the same after adjusting for key covariates namely age, duration of intervention and having another child during the follow-up period but were attenuated for maternal BMI and absolute fat mass when we also controlled for gestational weight gain and exclusive breastfeeding suggesting their mediating effects.

Major strengths of this study include the availability of prospectively collected data, double-blind RCT study design, large sample size with extensive follow-up, use of validated instruments, and availability of data on various confounding and mediating variables including breastfeeding status and duration, physical activity, dietary diversity, and reproductive history. A key strength is the assessment of body composition at 6 to 7 y PP using BIA. Our findings provide valuable insights for this study population whose mean percent body fat is higher than what is recommended as healthy (20%–25%) for females in this age group. The higher mean percent body fat for the MM group may be a concern for future chronic disease risk [53,54]. The differences in mean percent body fat especially among the females who were underweight at baseline may be clinically significant

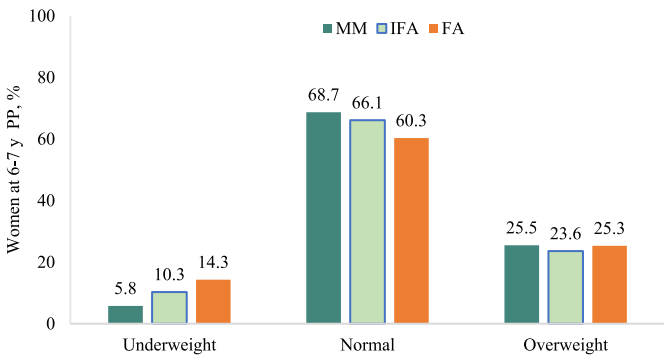


FIGURE 2. Prevalence of maternal underweight (BMI < 18.5 kg/m²), normal weight (18.5 ≤ BMI < 23 kg/m²), and overweight (BMI ≥ 23 kg/m²) at 6 to 7 y PP by treatment group¹ Abbreviations: ANOVA, analysis of variance; FA, folic acid; IFA, iron and folic acid; MM, multiple micronutrient; PP, postpartum. ¹Overall comparison ($P = 0.013$) and specific contrasts for prevalence of underweight: MM compared with IFA ($P = 0.007$), MM compared with FA ($P = 0.023$), and IFA compared with FA ($P = 1.00$) using ANOVA.

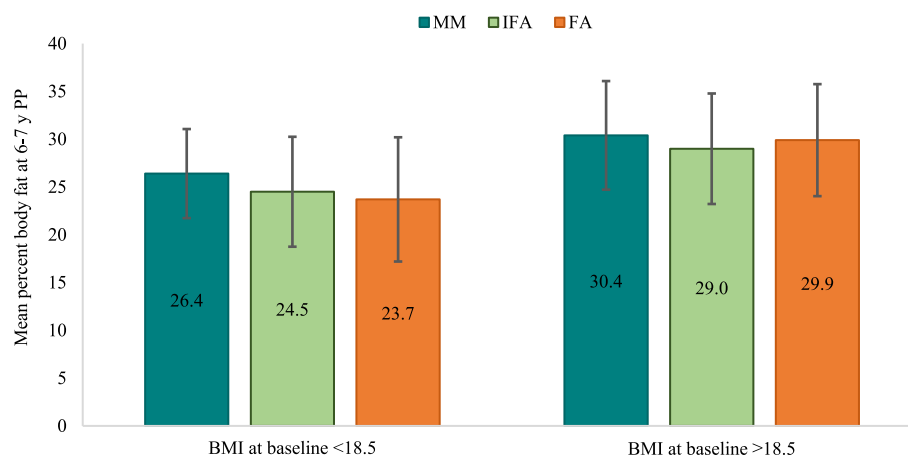


FIGURE 3. Interaction between maternal BMI at enrollment and treatment group for maternal percent fat at 6 to 7 y PP. ¹ $P = 0.078$ and 0.05 for interaction between maternal baseline nutritional status and treatment group before (unadjusted) and after adjusting for age, duration of intervention, having a child during the follow-up period, gestational weight gain during pregnancy, and exclusive breastfeeding, respectively. FA, folic acid; IFA, iron and folic acid; MM, multiple micronutrient; PP, postpartum.

(0.5 SD), although their values were within the healthy range (23%–26%). Another concern is the higher percent body fat values (29%–30%) that would be considered as unhealthy for females who were normal weight ($23 > \text{BMI} \geq 18.5$) at baseline [53,55,56].

A few important limitations should be considered in interpreting these findings. Although we have reliable data on factors such as breastfeeding practices and gestational weight gain, data on key determinants such as physical activity and dietary intakes were only measured at 6 to 7 y PP. It is possible that behaviors earlier in the PP period may have contributed to observed differences. Furthermore, we do not have data on body composition at baseline, the prevalence of comorbidities such as type 2 diabetes, as well as detailed reproductive history, gestational diabetes, or supplement use by females in the PP period. In addition, it is important to note that body composition assessed using the BIA and the manufacturer's algorithm may introduce some bias as previous studies suggest that some algorithms overestimate body fat mass in Asian populations [57,58]. However, it has been noted that any bias introduced by the algorithm is systematic and would not have a differential impact on any one of the study arms presented in this article. Thus, whatever bias may be introduced would not threaten the validity of the comparisons presented. Finally, the role of selection bias and lack of power need to be considered. Although baseline characteristics and treatment group assignment were similar across the analytic and excluded sample, the potential biases due to having older females and a higher proportion of females who had a child during the follow-up period need to be considered. Our final sample size also lacked power to detect small effect sizes <0.2 SD [30] combined with the small number of underweight females in the 3 treatment arms at 6 to 7 y PP.

To our knowledge, this is the first study to examine the long-term influence of preconception micronutrient interventions on maternal nutritional status and body composition. Results from this large RCT suggest that preconception micronutrient interventions may have long-term effects on maternal body composition. However, further research is needed using follow-

up data from other preconception and prenatal intervention studies to replicate findings and explore the biological mechanisms by which this may occur. The higher values of mean percentage body fat combined with observed increases in mean BMI and overweight over time are also very important and call for programs and policies that address the dual burden of malnutrition in countries like Vietnam. Although the benefits of MMs in reducing the prevalence of underweight is desirable in many low- to middle-income country settings, it is important to consider the potential risks of increased adiposity over time among underweight females in settings like Vietnam that are experiencing the nutrition transition and associated increases in NCDs [18,19].

Author contributions

The authors responsibilities were as follows – UR, MFY, PHN, MR: designed research; LQK, LMT, DJH: analyzed and interpreted data; UR, STW, MFY, RM, PHN, DJH: wrote article; UR: had primary responsibility for final content; all authors: critically read and provided feedback on manuscript; and all authors: read and approved the final manuscript.

Conflict of interest

The authors report no conflicts of interest.

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Data availability

Data described in the manuscript, code book, and analytic code will be made available upon request pending approval by the study team.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tjn.2024.02.024>.

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